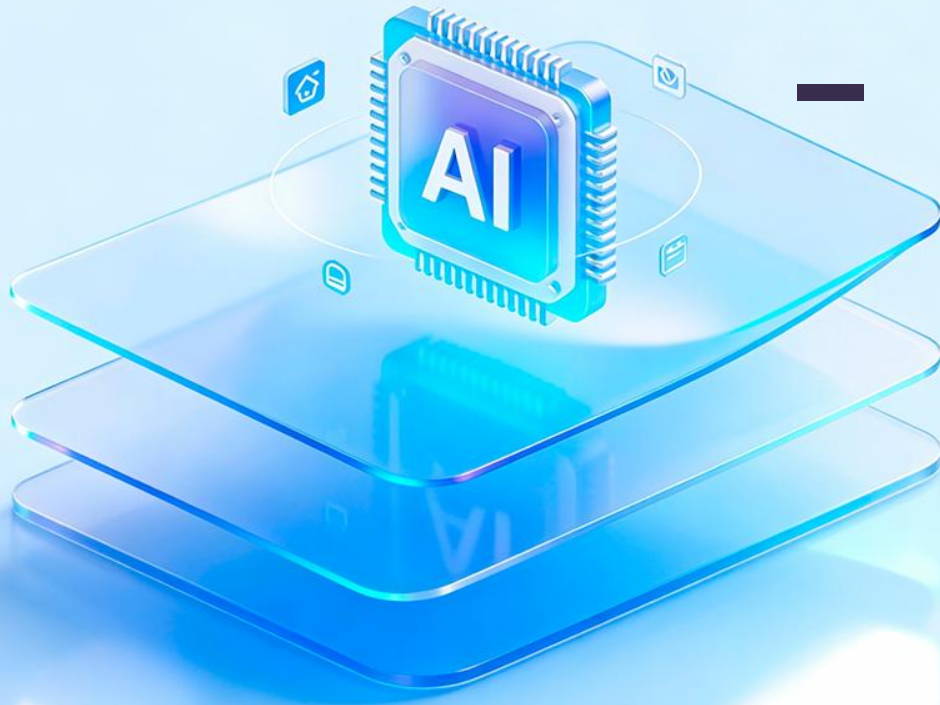


THE PRODUCER - CONSUMER PROBLEM



Course : Operating System

Class ID: 161859

Supervisor: Dr. Đỗ Quốc Huy

OUR TEAM

1. Trần Nguyễn Mỹ Anh - 20235474 - Leader
 2. Nguyễn Thu Huyền - 20235507
 3. Trần Lê Hạ Đan - 20235483
 4. Nguyễn Khánh Ly - 20235600
-

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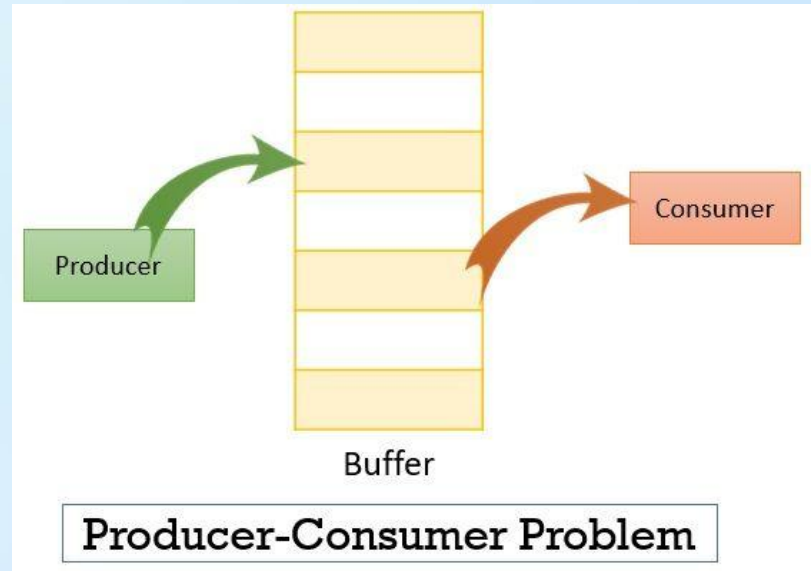
INTRODUCTION

OVERVIEW

Definition: A classical synchronization problem involving multiple processes sharing a common resource.

Components:

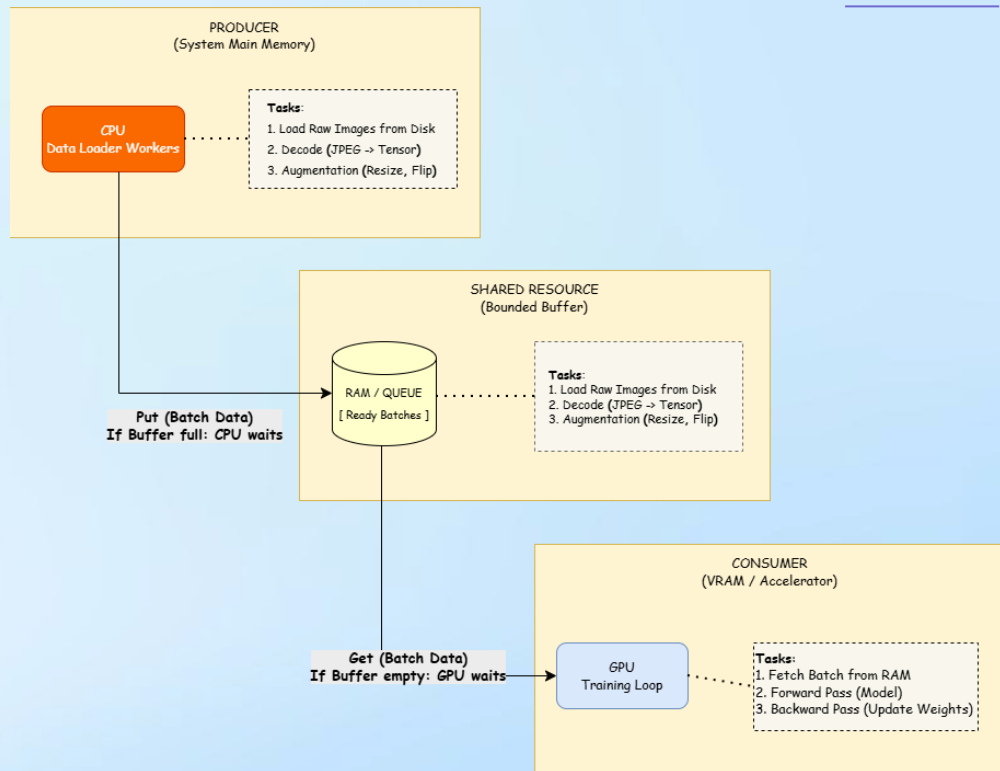
- **Producers:** Generate data items and place them in a shared buffer.
- **Consumers:** Remove and process items from the buffer.
- **Shared Buffer:** The critical resource where data is temporarily stored



OVERVIEW

AI/ML Context:

- **CPU (Producer):** Loads images, performs decompression, and data augmentation.
- **GPU (Consumer):** Retrieves batches for model training.
- **Challenge:** Poor coordination leads to the GPU remaining idle, wasting computational resources



PROBLEM ANALYSIS



PROBLEM STATEMENT

- **Key Components:** The system consists of Producers (responsible for generating data items) and Consumers (who process or use those items).
- **Communication Mechanism:** Interaction occurs through a shared buffer that temporarily stores items until they are consumed.
- **Bounded-Buffer Constraints:** In this variant, buffer usage is limited by its capacity.
 - **Overflow:** Producers must wait when the buffer is full.
 - **Underflow:** Consumers must wait when the buffer is empty.
- **Mutual Exclusion:** This is a critical requirement ensuring that only one process or thread modifies the buffer at a time to prevent data corruption

POTENTIAL ISSUES

Concurrency Issues:

- **Race Conditions:** Multiple threads modifying shared variables (counters/indices) simultaneously.
- **Data Inconsistency:** Items may be overwritten or lost.
- **Deadlock & Starvation:** Threads waiting indefinitely for conditions that never become true

Performance issues:

- **CPU Utilization:** Inefficient techniques like Busy Waiting waste processing resources because threads continuously check conditions instead of blocking.
- **Synchronization Overhead:** Frequent context switches and unnecessary wake-ups can significantly degrade system performance.
- **Scalability:** Performance may decrease as the number of producers and consumers increases

POTENTIAL ISSUES

Problem Objectives: A correct and optimized solution must satisfy the following criteria:

1. **Ensure Mutual Exclusion** during all buffer access operations.
2. **Prevent Buffer Overflow and Underflow** through proper coordination.
3. **Avoid Deadlock and Starvation** to ensure system progress.
4. **Optimize CPU Usage** by using blocking mechanisms instead of busy waiting.
5. **Support Scalability** by allowing multiple concurrent producers and consumers.

SYNCHRONIZATION MECHANISMS



SYNCHRONIZATION MECHANISMS

Critical Section: The code segment accessing the shared resource.

Key Requirements:

1. **Mutual Exclusion:** Only one thread executes a critical section at a time.
2. **Condition Synchronization:** Coordination based on buffer state (Full/Empty)

THREE APPROACHES

Having established the core difficulties and the need for synchronization in the Producer-Consumer model, we now turn to evaluating three specific mechanisms: Busy Waiting, Blocking Synchronization (Condition Variables), and Semaphores.

1

Busy Waiting

2

**Blocking
Synchronization and
Condition Variables**

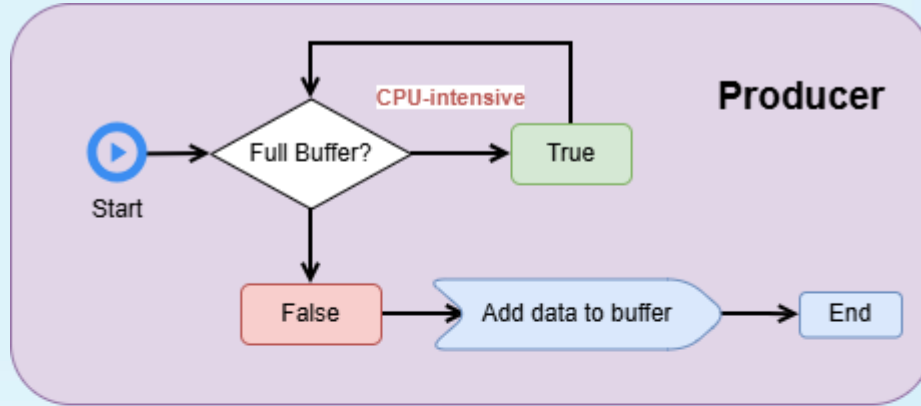
3

Semaphores

ALGORITHM DESIGN



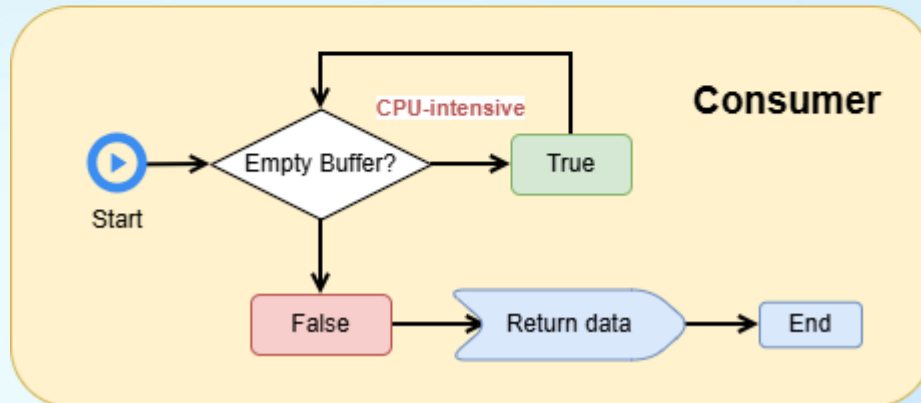
BUSY WAITING



Producer: Prior to production, it continuously queries:

“Is the buffer full?”

If it is full, the Producer waits and repeatedly re-queries until space becomes available.



Consumer: Prior to consumption, it continuously queries:

“Is the buffer empty?”

If the buffer is empty, the Consumer waits and repeatedly re-queries until data is available.

ALGORITHM

Input Data:

- Shared Resource: A simple Python list `buffer = []` acting as the storage.
- Constraints: `BUFFER_SIZE = 5`
- Control flags: A global variable **count** is used to check the size to determine the current state of the Buffer.

Output: Console logs

To implement this method:

- Utilize an **empty while** loop to hold the process execution flow while the condition remains unsatisfied

Algorithm 1: Busy Waiting

Input : Shared Buffer B with capacity N

Output : Data transfer from Producer to Consumer

// Global variable: `count` = current number of items

```

1 Function Producer():
2   while True do
3     item ← produce_item();
4     // Busy Wait: Loop
5     continuously while buffer is
6     full
7     while count ==  $N$  do
8       // Do nothing, just burn
9       CPU cycles
10    B[count] ← item;
11    count ← count + 1;
12
13 Function Consumer():
14   while True do
15     // Busy Wait: Loop
16     continuously while buffer is
17     empty
18     while count == 0 do
19       // Do nothing
20     item ← B[count - 1];
21     count ← count - 1;
22     process_item(item);
  
```


EVALUATION

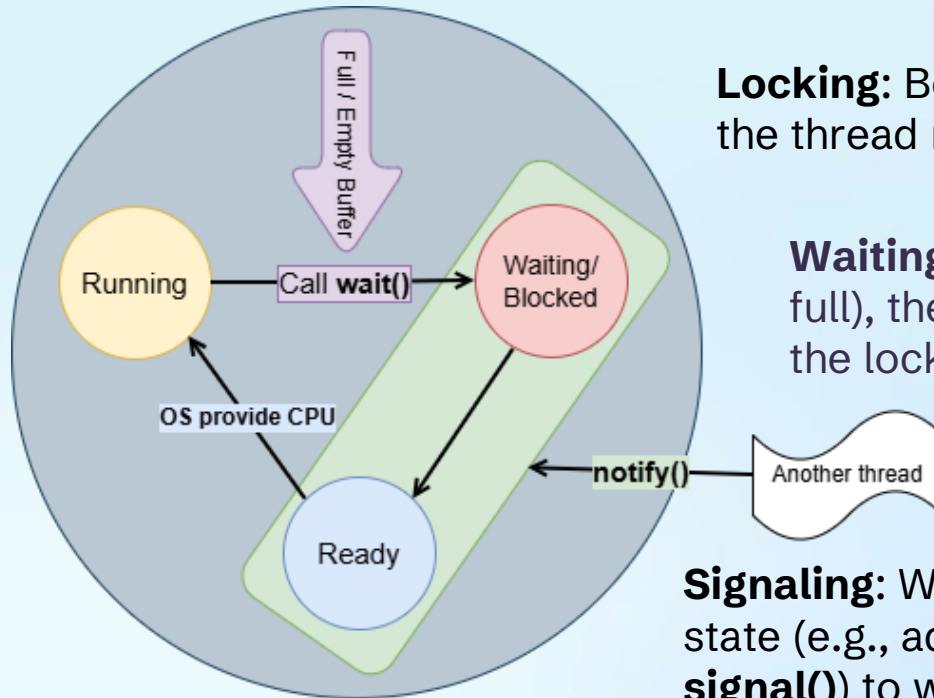
- **Strengths**

- Simple and easy to implement
- No context switch overhead (can be fast in rare short-wait cases)

- **Weaknesses**

- Severe CPU waste due to constant polling
- Race conditions may cause data corruption
- Poor scalability as producers/consumers increase

SLEEP & WAKEUP MECHANISM (CONDITION VARIABLES)



Locking: Before accessing the shared buffer, the thread must acquire a lock.

Waiting: If the condition is not met (e.g., buffer is full), the thread calls **wait()**. This action releases the lock and pauses the thread's execution.

Signaling: When a thread changes the buffer's state (e.g., adds an item), it calls **notify()** (or **signal()**) to wake up sleeping threads.

ALGORITHM

Input data:

- Synchronization Object: **threading.Condition()**. This object acts as both a Lock (Mutex) and a waiting room for threads.
- Shared Resource: The same **buffer** list with size 5.

Program output:

- [Producer] Buffer Full. Waiting... (Thread stops here).
- [Consumer] Consumed item. Notifying...
- [Producer] Woke up. Produced item.

Algorithm 2: Sleep & Wakeup (Condition Variables)

Input : Buffer B , Monitor Lock L ,
Condition Variable CV

```
1 Function Producer():  
2   while True do  
3      $item \leftarrow \text{produce\_item}()$ ;  
4     AcquireLock( $L$ );  
5     // Enter Critical Section  
6     while  $\text{size}(B) == N$  do  
7       // Release lock  $L$  and go to  
8       sleep  
9       Wait( $CV$ );  
10    push( $B, item$ );  
11    // Wake up sleeping Consumer  
12    Notify( $CV$ );  
13    ReleaseLock( $L$ );  
14  
15 Function Consumer():  
16   while True do  
17     AcquireLock( $L$ );  
18     // Enter Critical Section  
19     while  $\text{size}(B) == 0$  do  
20       // Release lock  $L$  and go to  
21       sleep  
22       Wait( $CV$ );  
23      $item \leftarrow \text{pop}(B)$ ;  
24     // Wake up sleeping Producer  
25     Notify( $CV$ );  
26     ReleaseLock( $L$ );  
27     process_item( $item$ );
```

ALGORITHM

- **Wait(condition):** This is the key difference from Method 1. Instead of looping, the thread stops execution here, saving CPU resources.
- **Signal(condition):** This command ensures that if the other party is sleeping, they will be alerted that the state has changed (e.g., from Empty to Not Empty).

Algorithm 2: Sleep & Wakeup
(Condition Variables)**Input** : Buffer B , Monitor Lock L ,
Condition Variable CV

```
1 Function Producer():  
2   while True do  
3      $item \leftarrow \text{produce\_item}()$ ;  
4     AcquireLock( $L$ );  
5     // Enter Critical Section  
6     while  $\text{size}(B) == N$  do  
7       // Release lock  $L$  and go to  
8       sleep  
9       Wait( $CV$ );  
10    push( $B$ ,  $item$ );  
11    // Wake up sleeping Consumer  
12    Notify( $CV$ );  
13    ReleaseLock( $L$ );  
14  
15 Function Consumer():  
16   while True do  
17     AcquireLock( $L$ );  
18     // Enter Critical Section  
19     while  $\text{size}(B) == 0$  do  
20       // Release lock  $L$  and go to  
21       sleep  
22       Wait( $CV$ );  
23      $item \leftarrow \text{pop}(B)$ ;  
24     // Wake up sleeping Producer  
25     Notify( $CV$ );  
26     ReleaseLock( $L$ );  
27     process_item( $item$ );
```

EVALUATION

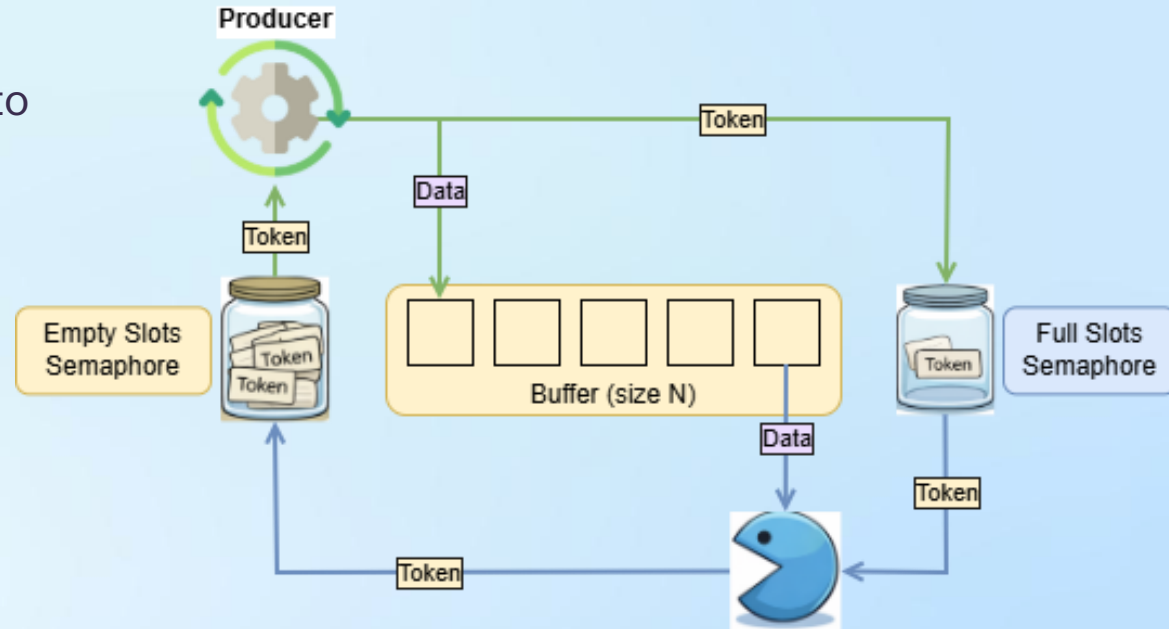
- **Strengths**
 - **CPU efficient:** no busy waiting while threads are blocked
 - **Thread-safe:** mutex/locks prevent race conditions
- **Weaknesses**
 - Context switch overhead when blocking/waking threads
 - More complex logic
 - Wake up without a signal → Require condition checks inside **while**, not **if**

SEMAPHORE & MUTEX

Idea: An optimized solution focusing on managing permits to access resources

Primitives Used:

- **empty_slots:** Counting semaphore initialized to Buffer Size
- **full_slots:** Counting semaphore initialized to 0
- **mutex:** Binary semaphore/lock initialized to 1 for mutual exclusion



Operations: P (Wait) decrements the count/blocks; V (Signal) increments/wakes

ALGORITHM

Input data:

- **empty_slots** : A *threading.Semaphore* initialized to BUFFER_SIZE (5). Represents the number of free spaces.
- **full_slots**: A *threading.Semaphore* initialized to 0. Represents the number of items ready for consumption.
- **mutex**: A *threading.Lock()* to protect the data integrity of the list.

Program output:

If the Producer is faster, the logs will show [Producer] Waiting for empty slot... and it will strictly wait until a [Consumer] Consumed... log appears

Algorithm 3: Semaphores (Optimized Solution)

Input : Semaphores: *Empty* (init N),
Full (init 0), *Mutex* (init 1)

1 **Function** Producer():

2 **while** *True* **do**

3 $item \leftarrow \text{produce_item}()$;
 // Step 1: Wait for an empty
 slot

4 Wait(*Empty*);
 // Step 2: Enter Critical
 Section (Exclusive Access)

5 Wait(*Mutex*);
6 add_to_buffer($item$);
7 Signal(*Mutex*);
 // Step 3: Signal that a new
 item is available

8 Signal(*Full*);

9 **Function** Consumer():

10 **while** *True* **do**

 // Step 1: Wait for available
 data

11 Wait(*Full*);
 // Step 2: Enter Critical
 Section

12 Wait(*Mutex*);
13 $item \leftarrow \text{remove_from_buffer}()$;
14 Signal(*Mutex*);
 // Step 3: Signal that a slot is
 now empty

15 Signal(*Empty*);
16 process_item($item$);

ALGORITHM

- **Separation of Concerns:** The **Wait(empty_slots)** and **Signal(full_slots)** pair in the Producer manages the flow control, while **Wait(mutex)** handles data integrity.
- **No spurious wakeups:** Unlike Method 2, Semaphores generally do not suffer from spurious wakeups, making the logic cleaner (no While loop needed for the lock itself, though the semaphore handles the blocking).

Algorithm 3: Semaphores (Optimized Solution)

Input : Semaphores: *Empty* (init N),
Full (init 0), *Mutex* (init 1)

```
1 Function Producer():  
2   while True do  
3     item  $\leftarrow$  produce_item();  
     // Step 1: Wait for an empty  
     slot  
4     Wait(Empty);  
     // Step 2: Enter Critical  
     Section (Exclusive Access)  
5     Wait(Mutex);  
6     add_to_buffer(item);  
7     Signal(Mutex);  
     // Step 3: Signal that a new  
     item is available  
8     Signal(Full);  
  
9 Function Consumer():  
10  while True do  
    // Step 1: Wait for available  
    data  
11  Wait(Full);  
    // Step 2: Enter Critical  
    Section  
12  Wait(Mutex);  
13  item  $\leftarrow$  remove_from_buffer();  
14  Signal(Mutex);  
    // Step 3: Signal that a slot is  
    now empty  
15  Signal(Empty);  
16  process_item(item);
```


EVALUATION

- **Strengths**

- Highly scalable: supports multiple Producers and Consumers efficiently
- Deadlock-safe with correct semaphore ordering
- Industry-standard model used in message queues and AI data pipelines

- **Weaknesses**

- Higher implementation complexity
- Semaphore misordering may cause deadlock
- Debugging synchronization errors is difficult

TEST CASES



SCENARIO 1: BUFFER OVERFLOW

- **Configuration:**
 - **Buffer size** = 2.
 - **Producer sleep time** = 0.1s.
 - **Consumer sleep time** = 1.0s.
- **Expected behavior:** The Producer should fill the buffer quickly and then enter a “Waiting” state. It must not overwrite existing data.

```
--- DEMO METHOD 1: BUSY WAITING & HIGH CPU ---  
Program will run in 15 seconds...  
  
[MONITOR] Start CPU Monitoring... (Wait 1-2 seconds for stable readings)  
[SYSTEM WARNING] High CPU Usage: 69.2% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.5% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.4% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 86.1% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.3% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.5% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 83.6% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.8% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.5% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.6% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.6% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.8% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 89.6% (Busy Waiting detected!)  
[SYSTEM WARNING] High CPU Usage: 90.1% (Busy Waiting detected!)  
  
--- STOP PROGRAM ---  
[SYSTEM WARNING] High CPU Usage: 74.8% (Busy Waiting detected!)
```

CPU usage - Method 1

Busy Waiting (Method 1): CPU usage spiked to 90.1% as the Producer loop checked the condition continuously

CPU usage - Method 3

Semaphores :

The Producer paused
efficiently.

```
--- DEMO METHOD 3: SEMAPHORES & LOW CPU ---  
Program will run for 15 seconds...  
  
[MONITOR] Start CPU Monitoring... (Wait 1-2 seconds for stable readings)  
[SYSTEM] Excellent! CPU Usage: 0.1% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.3% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.0% (Optimized)  
[SYSTEM] Excellent! CPU Usage: 0.2% (Optimized)  
  
--- STOP PROGRAM ---  
[SYSTEM] Excellent! CPU Usage: 0.1% (Optimized)
```

Console log showed: *[Producer] Waiting for empty slot....*

SCENARIO 2: BUFFER UNDERFLOW

- **Configuration:**
 - Buffer size $N = 5$.
 - Producer sleep time = 1.0s.
 - Consumer sleep time = 0.1s.
- **Expected behavior:** The Consumer should empty the buffer and then wait. It must not crash or retrieve None/garbage data.

SCENARIO 2: BUFFER UNDERFLOW

```
--- TEST SCENARIO 2: BUFFER UNDERFLOW (CONSUMER > PRODUCER) ---
Config: Buffer=5, Producer Sleep=1.0s, Consumer Sleep=0.1s
-----
[13:28:20] [Producer] Added 42.      Buffer: [42] (Len: 1)
[13:28:20] [Consumer] Waiting for data...
[13:28:20] [Consumer] Consumed 42.   Buffer: [] (Len: 0)
[13:28:20] [Consumer] Waiting for data...
[13:28:21] [Producer] Added 45.      Buffer: [45] (Len: 1)
[13:28:21] [Consumer] Consumed 45.   Buffer: [] (Len: 0)
[13:28:21] [Consumer] Waiting for data...
[13:28:22] [Producer] Added 51.      Buffer: [51] (Len: 1)
[13:28:22] [Consumer] Consumed 51.   Buffer: [] (Len: 0)
[13:28:22] [Consumer] Waiting for data...
[13:28:23] [Producer] Added 58.      Buffer: [58] (Len: 1)
[13:28:23] [Consumer] Consumed 58.   Buffer: [] (Len: 0)
[13:28:23] [Consumer] Waiting for data...
[13:28:24] [Producer] Added 28.      Buffer: [28] (Len: 1)
[13:28:24] [Consumer] Consumed 28.   Buffer: [] (Len: 0)
[13:28:24] [Consumer] Waiting for data...
[13:28:25] [Producer] Added 17.      Buffer: [17] (Len: 1)
[13:28:25] [Consumer] Consumed 17.   Buffer: [] (Len: 0)
[13:28:25] [Consumer] Waiting for data...
[13:28:26] [Producer] Added 70.      Buffer: [70] (Len: 1)
[13:28:26] [Consumer] Consumed 70.   Buffer: [] (Len: 0)
[13:28:26] [Consumer] Waiting for data...
[13:28:27] [Producer] Added 100.     Buffer: [100] (Len: 1)
[13:28:27] [Consumer] Consumed 100.  Buffer: [] (Len: 0)
[13:28:27] [Consumer] Waiting for data...
[13:28:28] [Producer] Added 5.       Buffer: [5] (Len: 1)
[13:28:28] [Consumer] Consumed 5.     Buffer: [] (Len: 0)
[13:28:28] [Consumer] Waiting for data...
[13:28:29] [Producer] Added 71.      Buffer: [71] (Len: 1)
[13:28:29] [Consumer] Consumed 71.   Buffer: [] (Len: 0)
[13:28:29] [Consumer] Waiting for data...
```

```
--- STOPPING TEST ---
```

Result : The Consumer paused correctly.

Console log showed: *[Consumer] Waiting for data....*

SCENARIO 3: CONCURRENCY & DATA INTEGRITY

- **Configuration:** 5 Producers and 5 Consumers operating simultaneously.
- **Expected behavior:** All items produced must be consumed exactly once. No items should be lost or duplicated.


```
--- TIME'S UP! STOPPING PRODUCERS... ---  
-----  
FINAL RESULT (DATA INTEGRITY CHECK):  
Total Items Produced: 8992  
Total Items Consumed: 8992  
  
>>> SUCCESS: PERFECT MATCH! NO DATA LOSS/DUPLICATION. <<<
```

Result:

Using method 3 (Semaphores + Mutex), the total count of produced items matched the total count of consumed items perfectly after a 60-second run

- **The most robust and suitable technique for real-world systems, offering the best balance of performance and scalability**

SCENARIO 4: MONITOR LOGIC VERIFICATION (SLEEP & WAKEUP)

- **Objective:** Verify that threads correctly enter the "Waiting" state when conditions are not met and resume execution only upon notification.
- **Configuration:**
 - Buffer Size: $N=2$.
 - Producer speed > Consumer speed (to force Buffer Full).
- **Expected behavior:** The Producer log must explicitly show a transition from "Going to sleep" to "Woke up" only after a Consumer action.

SCENARIO 4

```
--- TEST METHOD 2: SLEEP & WAKEUP (MONITOR) ---
[13:42:13] [Producer] Added 27. Buffer: [27]
[13:42:13] [Consumer] Took 27. Buffer: []
[13:42:14] [Producer] Added 79. Buffer: [79]
[13:42:14] [Consumer] Took 79. Buffer: []
[13:42:14] [Producer] Added 100. Buffer: [100]
[13:42:15] [Producer] Added 22. Buffer: [100, 22]
[13:42:15] [Consumer] Took 100. Buffer: [22]
[13:42:15] [Producer] Added 73. Buffer: [22, 73]
[13:42:16] [Producer] Buffer FULL. Going to sleep... (Wait)
[13:42:16] [Consumer] Took 22. Buffer: [73]
[13:42:16] [Producer] Woke up! (Notified)
[13:42:16] [Producer] Added 68. Buffer: [73, 68]
[13:42:17] [Producer] Buffer FULL. Going to sleep... (Wait)
[13:42:17] [Consumer] Took 73. Buffer: [68]
[13:42:17] [Producer] Woke up! (Notified)
[13:42:17] [Producer] Added 10. Buffer: [68, 10]
[13:42:18] [Producer] Buffer FULL. Going to sleep... (Wait)
[13:42:18] [Consumer] Took 68. Buffer: [10]
[13:42:18] [Producer] Woke up! (Notified)
[13:42:18] [Producer] Added 66. Buffer: [10, 66]
[13:42:19] [Producer] Buffer FULL. Going to sleep... (Wait)
[13:42:19] [Consumer] Took 10. Buffer: [66]
[13:42:19] [Producer] Woke up! (Notified)
[13:42:19] [Producer] Added 75. Buffer: [66, 75]
[13:42:20] [Producer] Buffer FULL. Going to sleep... (Wait)
[13:42:20] [Consumer] Took 66. Buffer: [75]
[13:42:20] [Producer] Woke up! (Notified)
[13:42:20] [Producer] Added 48. Buffer: [75, 48]
[13:42:21] [Producer] Buffer FULL. Going to sleep... (Wait)
[13:42:21] [Consumer] Took 75. Buffer: [48]
[13:42:21] [Producer] Woke up! (Notified)
[13:42:21] [Producer] Added 76. Buffer: [48, 76]
[13:42:22] [Producer] Buffer FULL. Going to sleep... (Wait)
[13:42:22] [Consumer] Took 48. Buffer: [76]
[13:42:22] [Producer] Woke up! (Notified)
[13:42:22] [Producer] Added 77. Buffer: [76, 77]
[13:42:23] [Producer] Buffer FULL. Going to sleep... (Wait)
[13:42:23] [Producer] Woke up! (Notified)
[13:42:23] [Producer] Buffer FULL. Going to sleep... (Wait)
```

Result: The console log confirmed the behavior. The Producer paused when the buffer reached size 2 and resumed immediately after the Consumer removed an item

**THANK
YOU**

**DO YOU HAVE ANY
QUESTIONS?**